



Insights into research on blockchain for smart contracts: a bibliometric analysis

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Abstract

Over the past few years, blockchain technology has gained significant attention. This surge in popularity can be attributed to the emergence of cryptocurrencies and the development of smart contracts. Cryptocurrency is a digital currency that eliminates the problem of double spending. Cryptocurrencies like Bitcoin, Ethereum, Litecoin, Stellar, Zcash, Maker, Aave, etc. become popular and are preferred for money transfers. Smart contracts are the next popular technology on the blockchain after cryptocurrency. It can be considered a piece of code that can execute automatically when the predefined conditions are fulfilled. Researchers believe that the potential of blockchain with smart contracts is only in its initial stages and that its true potential has yet to be fully discovered. Hence, an extensive bibliometric analysis is conducted to understand blockchain trends for smart contracts and to give future directions in this field. For this analysis, various steps are followed, starting with formulating the research question, defining the scope of our research, extracting and analyzing data, answering the research question, and finally, drawing a conclusion. This research paper will be fruitful for scholars and researchers, providing an extensive statistical and network analysis of extracted smart contracts publications.

Keywords Scientometrics · Blockchain · Smart contracts · Crypto contract · Bibliometric analysis · Scopus · VOSViewer

1 Introduction

Blockchain Technology (BT) was revealed by Satoshi Nakamoto in 2008 with his white paper, "Bitcoin: A Peer-to-Peer Electronic Cash System" [1]. This paper was the mathematical foundation of Bitcoin, the first blockchain-based cryptocurrency. Before the advent of Bitcoin, third parties like banks and government organizations were utilized as

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middlemen to make financial transactions. But Bitcoin completely changed the scenario. As of now, transactions can be done without the involvement of any mediator. This is possible only because of the blockchain's decentralized, transparent, and immutable nature to record transactions.

Satoshi Nakamoto mentioned in his whitepaper that blocks are chained, which led to the name "Blockchain." Each block of the blockchain stores the transaction records. Satoshi Nakamoto created a hash value using the SHA-256 algorithm to identify each blockchain block uniquely. This method of linking data with a hash algorithm is also mentioned in the timestamping of digital documents [2]. Although Blockchain Technology began in 2008, it got public attention when it appeared on the front page of a magazine called "The Economist" in 2015, with the title "The Trust Machine: How the Technology Behind Bitcoin Could Change the World" [3]. Smart contracts represent a notable use of blockchain technology. They are essentially programs embedded within the blockchain that are automatically executed once their specific criteria are met [4]. It was developed by Nick Szabo [5] in the 1990s. Since then, much research has been conducted on smart contracts and blockchain technology. Hence, the focus of this study is exploring blockchain-based smart contracts.

Bibliometric analysis is one of the most popular literature review methods [6]. Various researchers have recently explored blockchain technology using the bibliometric method. In one study, Sugandh et al. [8] studied the utilization of blockchain in food and agriculture. Another study conducted by Marques et al. [7] identifies the usage of blockchain technology in higher education. Nasir et al. [8] study explores cryptocurrencies and their implications on the security and privacy of blockchain technologies. Mubigmann et al. [9] perform citation and co-citation analysis on Logistics and Supply Chain Management using blockchain. Table 1 presents the summary of the previous literature using Bibliometric Analysis.

Although researchers in the field of blockchain technology widely utilize Bibliometric Analysis. However, the study focused on smart contracts is addressed only by Lennart Ante [10] by considering 468 peer-reviewed publications and their references extracted from the Web of Science. Hence, for more exploration of smart contracts on current data, this study on blockchain-based smart contracts is conducted.

The scope of the study on "Blockchain for Smart Contracts" is as follows:

- a) This analysis includes 10,862 publications from the last ten years.
- b) This analysis assesses different research efforts in blockchain for smart contracts through various documents.
- c) This study investigates the various extracted documents in two ways:
 - a. Statistical analysis, and
 - b. Network analysis.

The structure of the research paper is divided into several key sections. Initially, the Background section thoroughly examines Blockchain Technology and Smart Contracts. This is followed by the Research Methodology section, which outlines the development of research questions and the data extraction and analysis approaches. Subsequently, the Discussion section delves into the findings from various analyses and addresses the research questions. This leads to an examination of the practical implications and limitations of the study. The paper concludes with a Conclusion summarizing the study's findings and contributions.

Table 1 Summary of previous literature using bibliometric analysis

Paper	Objective of study	Search term	Database used	No. of articles	Software used
[11]	The study focuses on the evolution of blockchain and its research areas	Blockchain*	Web of Science	2993	SciMAT
[12]	The objective is to explore how blockchain is applied in food and agriculture	Agriculture, Blockchain, smart agriculture, E-agriculture, precision agriculture, food security, food safety, agro-food supply, agriculture supply chain, food supply chain	IEEE Xplore, ACM, PubMed, Scopus, Web of Science	173	VOSViewer
[7]	The focus is to implement blockchain in Higher Education	Blockchain, Higher Education, Digital	Scopus	61	VOSViewer
[8]	The topics considered are cryptocurrency structure, implications, privacy, and data security	Blockchain, Bitcoin, Cryptocurrency, Digital Currency, Technology	Web of Sciences, Scopus, Elsevier, Emerald, Google Scholar	1965	Bibliometrix, R package, and VOSViewer
[9]	The study presents a bibliometric review of logistics and supply chains by considering publications from different sources	Supply Chain, Transport, Logistics, Blockchain, Distributed Ledger Technology	IEEE Xplore, Google Scholar, Springer, Emerald Insight, Taylor & Francis, Scopus, Science Direct, SSRN, and WoS	613	Gephi
[10]	The study explores the smart contract trends	Smart Contract*	Web of Science	468	VOSViewer

2 Background

Smart contracts are executed on Ethereum, which relies on blockchain. One must have some background knowledge of blockchain to understand and develop smart contracts. Hence, a concise description of Blockchain Technology and Smart Contracts is discussed in this section.

2.1 Blockchain technology

With Bitcoin's emergence, Blockchain Technology has drawn increased focus from the academic and industrial sectors. Its characteristics like decentralization, immutability, persistency, security, capacity, anonymity, and auditability make it different and stand out from other technologies with the potential to change many industries and businesses. Various researchers and industrialists define blockchain in their own terms. One of the ways of defining blockchain is as a method of storing unviable information to hack, change, or cheat the data.

In Simple words, one can understand blockchain as a group of computers connected, making the entire network decentralized [13]. As per Iansiti et al. [14], "*Blockchain is an open, distributed ledger that can record transactions between two parties efficiently and in a verifiable and permanent way.*" Crosby et al. [15] defined blockchain as "*a distributed database of records or public ledger of all transactions or digital events that have been executed and shared among participating parties.*" Another definition of blockchain by Xu et al. [16] is, "*Blockchain is a distributed ledger that is structured into a linked list of blocks. Each block contains an ordered set of transactions. Typical solutions use cryptographic hashes to secure the link from a block to its predecessor.*"

According to the author's insight, blockchain has no formal definition. But the definition presented by Vitalik Buterin [17]: "*A blockchain is a magic computer that anyone can upload programs to and leave the programs to self-execute, where the current and all previous states of every program are always publicly visible, and which carries a solid cryptoeconomically secured guarantee that programs running on the chain will continue to execute in exactly the way that the blockchain protocol specifies,*" includes all the characteristics of blockchain and can be thought of as a better way of defining blockchain than others.

2.2 Architecture of blockchain

A Blockchain is a sequence of blocks connected, as illustrated in Fig. 1. Every block of blockchain stores some details for the unique identification of each block. The details in the block are the hash of the block, the hash of the preceding block, a time stamp, a nonce, the Merkle root, and the data from transactions [18]. The initial block in the blockchain is known as the Genesis block or Block 0. This block is distinct because it has no previous blocks, and subsequent blocks are appended to the blockchain following Block 0.

1. Hash: A hash of a block is an alphanumeric number that is used to identify each block inside the blockchain network uniquely.
2. Hash of the preceding block: It is the hash of the prior block of the blockchain.
3. Timestamp: A timestamp is when a transaction occurs per the universal time of January 1, 1970.

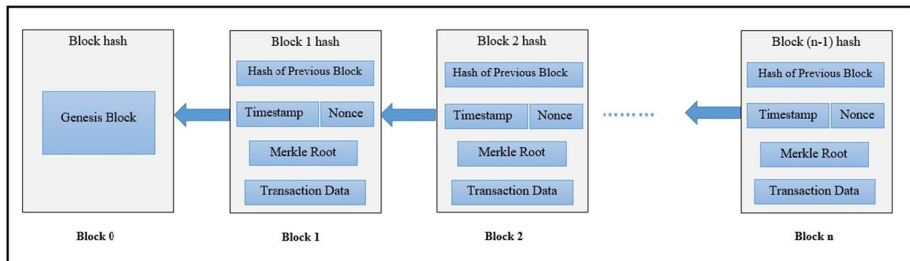


Fig. 1 Architecture of blockchain

4. Nonce: A random number that changes the hash value.
5. Merkle Root: It represents the combined hash of all transaction hashes in a blockchain block.
6. Transaction data: It is the information that is securely recorded for a transaction, such as sender, receiver, amount, and timestamp.

Blockchain is an emerging technology with applications in the financial sector, supply chain, internet of Things, healthcare, governmental services, and smart contracts. In the financial industry [19], blockchain has use cases in asset management [20], capital markets [21], banking and lending [22], payments and remittances [23], trade finance [24], and insurance [25]. In the supply chain [26], blockchain can be used for traceability, product authentication, quality control, and inventory management. In healthcare [27], blockchain can be used for Electronic Medical Records (EMR) [28], Drug/ Pharmaceutical supply chain [29], Biomedical Research and Education [30], Remote Patient Monitoring (RPM) [31], Health Insurance Claims [32], and Health Data Analytics (HDA) [33]. In the Internet of Things, blockchain has applications in Operational Maintenance, Freight transportation, and Smart homes [34]. In governmental services [35], blockchain can be used for Identity management and record keeping [36], Value registry [37], and voting system [38]. Blockchain can be used in smart contracts for cryptocurrency exchanges, insurance contracts, employee contracts, protecting copyrights, and supply-chain management [39].

2.3 Smart contracts

The blockchain block records certain transactions. With this, the idea of smart contracts came into play; why not store programs inside these blocks? These programs are further named as "smart contracts." In the 1990s, the first smart contract was developed by Nick Szabo [5].

After this, smart contracts started gaining attention from industrialists and researchers as, unlike traditional contracts, they do not need any third party, as illustrated in Fig. 2. Moreover, they are brief and document-based and can execute automatically when the conditions are met. In this way, smart contracts save money and time. Due to the benefits of smart contracts over traditional contracts, smart contracts became a popular research topic. The researchers believe that the potential of smart contracts is only in its initial stages, and their true potential has yet to be discovered.

A smart contract represents a straightforward piece of programming that utilizes blockchain technology to facilitate, execute, and enforce the terms of an agreement among parties

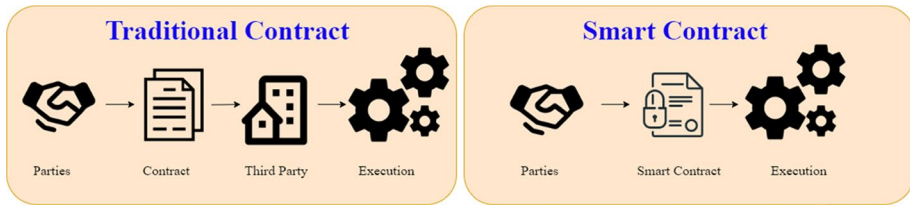


Fig. 2 Traditional contract vs smart contract

that do not necessarily trust one another. This mechanism awards digital assets to the parties involved once the conditions have been fulfilled [40].

Smart contracts work upon if-else conditions that help automate the transactions so that all parties are confident about the decision and do not need intermediaries. For the implementation of smart contracts, available platforms are Ethereum, Hyperledger Fabric, Nem, Stellar, Waves, DocuSign, Solana, and Velas. But Ethereum is the most popular of all. Ethereum's great popularity is due to its language having a Turing complete feature, which helps create more advanced and custom smart contracts.

2.3.1 Life cycle of smart contract

The smart contract development consists of three phases: contract generation, release, and execution.

- i. *Contract generation:* This stage includes multiple steps. In the first step, the parties involved in smart contracts clarify each other's rules and obligations. In the second step, they will write all these rules in the form of text, specifying the working conditions of the smart contract. In the third step, the specifications of the text are verified in the form of code. Finally, after the verification of the code, a standard code for the smart contract is generated. The two most important tasks done in the contract generation stage are the specification and verification of the contract. Experts and contract parties with related domain knowledge must create contract specifications. The contract verification is deployed on the Ethereum Virtual Machine [41] if the smart contract is written on Ethereum IDE. Contract verification is essential for checking the integrity of the code.
- ii. *Contract release:* The next stage is the contract release once the code is generated. In this case, the contract is dispersed to different nodes in the network in Peer-to-Peer mode. Each network node stores the contract in its memory and waits for the consensus mechanism to be fulfilled.
- iii. *Contract execution:* After contract release, the next stage is contract execution. At this stage, the contract cannot be changed. This stage is based on event trigger mechanisms [42]. Whenever the preset condition of the smart contract is fulfilled, the relevant condition will be triggered, and the smart contract will execute by responding to the preset response [43].

2.3.2 Types of smart contracts

Depending upon applications, smart contracts are subdivided into four classes, namely, (i) Smart legal contracts, (ii) Decentralized Applications (DApps), (iii) Distributed Autonomous Organisations (DAO), and (iv) Smart contracting devices.

According to [44], the smart legal contract is “a legally binding contract in which some or all of the contractual obligations are defined in and performed automatically by a computer program.” These contracts are more efficient and trustworthy than traditional legal contracts. DApps are those applications that execute on a P2P network of computers in place of one single computer [45]. Distributed Autonomous Organisation is an organization that works upon the shared missions of blockchain. No one has the authority to be a part of the organization without the permission of group members. All the decisions are made by proposals and voting to ensure transparency [46]. Smart contracts can be applied in any field to automate the work. For example, a smart contract can be used in a supply chain to automate the payment from the wallet when the products, goods, or services are delivered. In health care, smart contracts can be applied to store the complete history of patients and give access to the related parties like doctors or technicians to make changes required. In the next section, the research methodology is presented.

3 Research methodology

Research methodology is a systematic strategy to solve a particular research problem [47]. The primary steps are formulating research questions, data extraction, data analysis, performance analysis, and concluding the study by answering the research questions.

The study utilizes the methodology presented by Kitchenham et al. [48]. The detailed workflow is illustrated in Fig. 3. The research questions considered for this analysis are described below.

RQ1: What are the hot topics in Blockchain-based Smart Contracts?

RQ2: What is the present state of research on smart contract development?

RQ3: What are the applications of smart contracts?

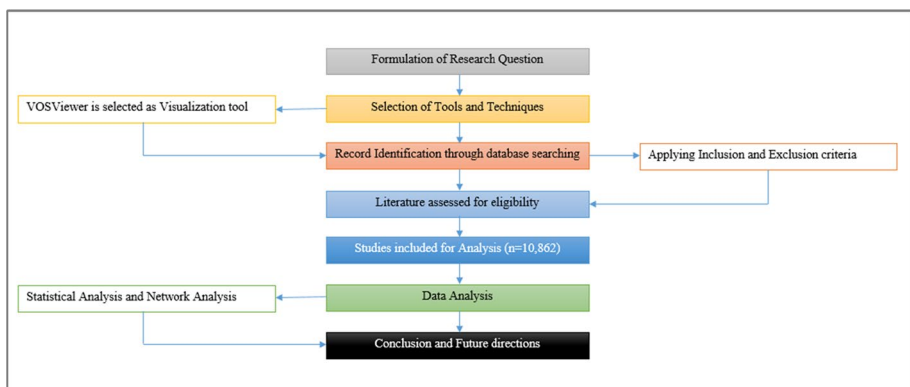


Fig. 3 Workflow of study (Kitchenham methodology)

RQ4: Which blockchains offer the best smart contract formulations?

RQ5: What are the challenges and limitations of smart contracts?

RQ6: What are the topics for investigation and the potential fields for upcoming research in smart contracts?

3.1 Bibliometric analysis

Bibliometric is a way of analyzing bibliometric publications like peer-reviewed journal publications, conference proceedings, books, review articles, periodicals, reports, and other research documents with the help of statistical techniques [49]. Using quantitative methods, the bibliometric method is globally used to present the association within research domains.

Hood et al. [50] state, "*Bibliometrics is the quantitative literature study reflected in bibliographies. Its task, immodestly enough, is to provide evolutionary models of science, technology, and scholarship.*" Researchers have conducted bibliometric analysis on different domains like Healthcare [50], Microplastics [52], Supply Chain Management [53], Finance [54], Energy [55], Environment [56], and many more.

The primary reasons behind the popularity of bibliometric analysis are the availability and accessibility of bibliometric software and the ability to handle huge amounts of data. Some of the popular software utilized for bibliometric analysis are BibExcel [51], Cite Space [52], Netdraw [53], Pajek [54], Sci2 [55], PoP [56] (Publish or Perish), R Studio [57], SciMAT [58], SITKIS [59], UCInet [60], and VOSviewer [61].

For this bibliometric analysis, VOSviewer [62] version 1.6.18 is utilized as software. It provides a very efficient method for analyzing the co-occurrences, co-citations, and bibliometric coupling. Moreover, it is more user-friendly and has excellent graphic quality with pre-programmed functions, making combining data from different database sources easy. VOS Viewer was originated in 2010 by "Nees Jan van Eck" and "Ludo Waltman" at "Leiden's University Centre for Science and Technology Studies." Since 2010, many publications have performed analyses with VOSviewer.

3.2 Data sources

Many databases are available for extracting bibliometric data, like Web of Science [63], Scopus [64], Scimago [65], Google Scholar [66], ScienceDirect [67], Dimensions [68], Emerald [69], IEEE Xplore [70], etc. All these databases include many high-impact and reputable research articles from Computer Science, Natural Science, Social Science, Art, and Humanity.

The success of any bibliometric analysis depends upon the satisfactory documentation of the research domain, which can be obtained through Scopus. As mentioned earlier, there are many advantages to choosing Scopus over databases. The first is that Scopus stands out from other databases with its high coverage of scientific literature, including journals, conference proceedings, and books from over 5000 publishers globally [71, 72]. Scopus offers valuable tools like author-affiliation linking and author profiling, which are crucial for studying international scientific migration [73]. Compared with other data sources like Web of Science and Dimensions, Scopus provides a valuable set of filters for making selections and ensures comprehensive coverage of scientific documents from various disciplines and document types [74]. Additionally, Scopus has a stringent content selection policy, advanced analysis tools, and real-world applications in national research assessments,

Table 2 List of keywords used for searching publications

S. No	Keywords
1	Blockchain Technology
2	Blockchain, Blockchains
3	Distributed ledger technology
4	A smart contract, smart contracts
5	Crypto contract, crypto contracts

Table 3 Trend of publications in different language

Language	No. of publications
English	11,342
Chinese	284
Spanish	29
German	22
Russian	15
Portuguese	12
Italian	7
Turkish	5
Korean	4
French	3
Croatian	2
Japanese	2
Catalan	2
Dutch	1
Polish	1
Total	11,728

university rankings, and research landscape studies [75]. Considering all these advantages of Scopus, it is utilized as the data extraction source for this study.

3.3 Data extraction

Data extraction involves identifying the keywords, creating the search query, and extracting the data in a suitable format. Different keywords related to the “Blockchain-based Smart Contracts” theme are identified in the first step. These keywords are enlisted in Table 2. In the second step, a search query is created using these keywords, as mentioned in Eq. 1. This search resulted in 11,728 publications published across the world. This search result applies with no restriction to the language and country. The Bibliometric data of author, citations, affiliations, journals, country, etc., is available for each publication.

Equation 1 initial search query (15 January 2023).

```
(TITLE – ABS – KEY((blockchain AND technology)
OR(blockchain*) OR(distributed AND ledger AND technology))
AND TITLE – ABS – KEY(smart AND contract*)OR(crypto AND contract*))
```

This search result includes publications in different languages like English, Chinese, Spanish, German, Russian, Portuguese, Italian, Turkish, Korean, and French. The trend of publications in other languages is presented in Table 3. The table shows that the English language has the highest publications and is preferred among the researchers for publishing research work. Also, Chinese is the second preferred language among the researchers.

In the third step, a set of inclusion and exclusion criteria (shown in Table 4) are applied to refine the results. After using the criteria, the final search query is shown in Eq. 2. In total, 10,862 total publications satisfy the criteria.

Equation 2 final search query (15 January 2023).

```
(TITLE – ABS – KEY(blockchain AND technology)
OR(blockchain*) OR(distributed AND ledger AND technology)
AND TITLE – ABS – KEY(smart AND contract*)
OR(crypto AND contract*)) AND (LIMIT – TO (LANGUAGE, eEnglishe))
AND (EXCLUDE (PUBYEAR, 2023)) AND (LIMIT – TO (PUBSTAGE, efinale))
AND (EXCLUDE (DOCTYPE, etbe) OR EXCLUDE (DOCTYPE, eere))
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4 Discussion

4.1 Data analysis

Data analysis is a systematic technique of investigating the data to draw conclusions or to increase knowledge [76]. Analysis of the extracted publications is performed based on keywords and citations. In this subsection, the keyword analysis is considered first, and then, the analysis is discussed based on several citations.

4.1.1 Analysis of search results based on keywords

The authors of different publications use different keywords. The top 20 keywords, according to their number of occurrences, are shown in Table 5. Blockchain is the keyword that has the maximum number of occurrences in different publications, followed by smart contracts. All these keywords are related to Blockchain and Smart Contracts.

Table 4 Inclusion–exclusion criteria for publications

Inclusion criteria	Publications having the Publication stage as Final are included Publications written in English are considered Publications of the last ten years, from 2013–2022, are included for analysis
Exclusion criteria	Document types like retracted and erratum are excluded

Table 5 Publications analysis depending upon top 20 keywords analysis

S. No	Keyword	No. of occurrences in publications
1	Blockchain	9228
2	Smart Contract	3949
3	Block-chain	3202
4	Smart Contracts	2078
5	Ethereum	1685
6	Internet Of Things	1434
7	Digital Storage	939
8	Network Security	880
9	Cryptography	834
10	Distributed Ledger	792
11	Security	688
12	Decentralized	687
13	Data Privacy	605
14	Information Management	560
15	Access Control	540
16	Cryptocurrency	525
17	Authentication	522
18	Bitcoin	519
19	Supply Chains	493
20	Privacy	472

4.1.2 Analysis of search results based on top-cited articles

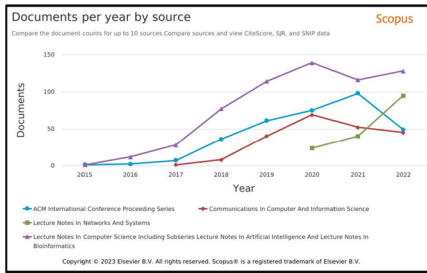
Among the selected publications for study, the top 15 articles have the highest number of citations retrieved. These publications are categorized by article type and country. In article type, the publications are under two categories: Review Article and Original Article. Of 15 publications, 9 are review articles, and six are original. The higher number of citations in review articles is because they act as a base for new research opportunities. The article “Blockchains and Smart Contracts for the Internet of Things” has the highest citation count of 2382. Based on Country, the Country of the first author is considered. The USA has three publications, followed by two publications from China. The detailed analysis of these publications is depicted in Table 6.

4.2 Statistical analysis

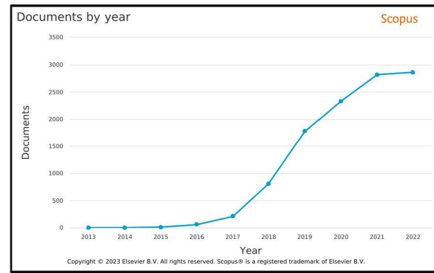
Statistical analysis is essential in bibliometrics as it aids the quantitative evaluation of scientific publications [92, 93]. It enables the measurement of various bibliometric performance indicators, such as publication growth, author collaboration, and keyword associations, by providing insights into research trends and developments [94, 95]. Overall, statistical analysis in bibliometrics aids in mapping research directions, identifying key contributors, and understanding the evolving landscape of various research

Table 6 Top 15 cited articles (Source: Scopus, Accessed Date: 15 January 2023)

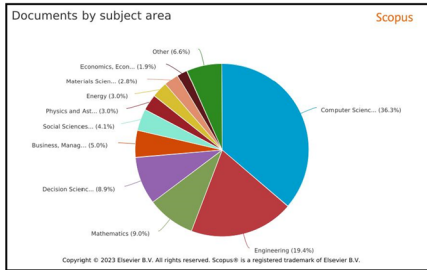
Rank	Article title	Article type	Year	Country	Citations
1	“Blockchains and Smart Contracts for the Internet of Things” [77]	Review Article	2016	USA	2382
2	“An Overview of Blockchain Technology: Architecture, Consensus, and Future Trends” [78]	Review Article	2017	China	1988
3	“Hyperledger Fabric: A Distributed Operating System for Permissioned Blockchains” [79]	Original Article	2018	New York	1509
4	“Blockchain challenges and opportunities: A survey” [80]	Review Article	2018	China	1390
5	“IoT security: Review, blockchain solutions, and open challenges” [81]	Review Article	2018	Pakistan	1343
6	“Hawk: The Blockchain Model of Cryptography and Privacy-Preserving Smart Contracts” [82]	Original Article	2016	USA	1339
7	“Blockchain technology and its relationships to sustainable supply chain management” [83]	Review Article	2019	USA	1113
8	“Where is current research on Blockchain technology?—A systematic review” [84]	Review Article	2016	Finland	1063
9	“Making smart contracts smarter” [85]	Original Article	2016	Singapore	1042
10	“Blockchain technology in the energy sector: A systematic review of challenges and opportunities” [86]	Review Article	2019	UK	1017
11	“Bitcoin and beyond: A technical survey on decentralized digital currencies” [87]	Review Article	2016	Germany	991
12	“On blockchain and its integration with IoT: Challenges and opportunities” [88]	Review Article	2018	Spain	910
13	“On the security and performance of Proof of Work blockchains” [89]	Original Article	2016	Switzerland	827
14	“Bitcoin-NG: A scalable blockchain protocol” [90]	Original Article	2016	New York	821
15	“An agri-food supply chain traceability system for China based on RFID & blockchain technology” [91]	Original Article	2016	Europe	788



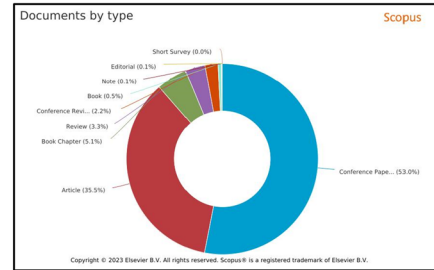
(i)



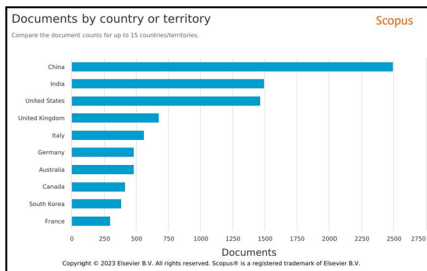
(ii)



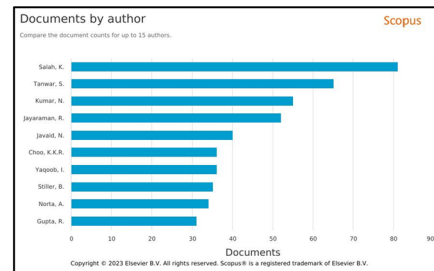
(iii)



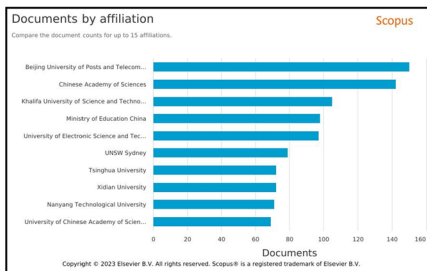
(iv)



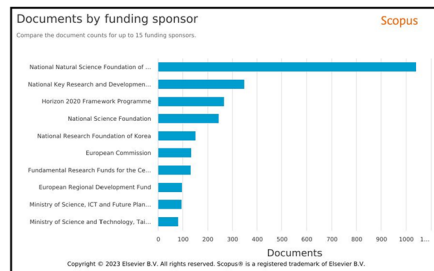
(v)



(vi)



(vii)



(viii)

Fig. 4 Documents analysis by (i) Source, (ii) Year, (iii) Subject area, (iv) Type, (v) Country/Territory, (vi) Author, (vii) Affiliation, and (viii) Funding agencies (Source: Scopus)

fields. The statistical analysis performed with the help of Scopus is document analysis by source, year, subject area, type, country, author, affiliation, and top funding agencies.

Table 7 Year-wise distribution of publications

Year	No. of publications
2022	2858
2021	2814
2020	2325
2019	1776
2018	808
2017	212
2016	58
2015	9
2014	1
2013	1
Total	10,862

- i. Documents Analysis by Source: Fig. 4i depicts the document analysis by source. It can be inferred that “Lecture Notes in Computer Science” has the highest publications, followed by publications in the “ACM International Conference Proceeding Series.” The possible reason for this trend is that blockchain is an emerging technology. Hence, researchers prefer to present publications at conferences to discuss and get research advice from the experts in this domain.
- ii. Documents Analysis by Year: The year-wise distribution of publications is listed in Table 7, and Fig. 4ii shows a graphical demonstration. It can be noticed that the year 2022 has the maximum publications, followed by 2021 and 2020, which means the popularity of this domain among researchers is increasing year over year.
- iii. Documents Analysis by Subject Area: On Analyzing the documents in terms of subject area as depicted in Fig. 4iii, it is noted that a maximum (around 36%) of publications have a subject area of Computer Science. The primary reason for this is the dual nature of blockchain. Blockchain is a data structure and a consensus network, which is the primary reason for its high number of publications in computer science.
- iv. Documents Analysis by Type: The publications retrieved from the Scopus database are from different sources, as depicted in Table 8. The maximum number of researchers prefer publishing at Conferences, followed by Articles. The possible reasons for maximum conference publications are: (1) they want to publish their novel ideas, (2)

Table 8 Distribution of type of publications

Type of document	No. of publications
Conference Paper	5761
Article	3857
Book Chapter	559
Review	363
Conference Review	242
Book	59
Note	10
Editorial	6
Short Survey	5
Total	10,862

they want to gain inputs from the experts in the domain, and (3) they want to enhance their conference paper for the journal. Further, Fig. 4iv shows the detailed analysis of different types of publications.

- v. Documents Analysis by Country/Territory: When Documents are analyzed based on Country or Territory, it is noted that China has the most publications, followed by India and the United States, as shown in Fig. 4v.
- vi. Documents Analysis by Authors: In this type of analysis, authors are selected based on their published publications. The top ten authors with the most significant count of documents are shown in Fig. 4vi. It is noticed that Salah, K. [96] is having a maximum count of 81 followed by Tanwar, S. [97].
- vii. Documents Analysis by Affiliation: Fig. 4vii shows the top ten Affiliation from the retrieved results. It is noted that “The Beijing University of Posts and Telecommunications” has the highest document count, followed by “The Chinese Academy of Sciences” and “Khalifa University of Science and Technology”.
- viii. Documents Analysis by top funding agencies: In Fig. 4viii, the top ten funding agencies are considered. The “National Natural Science Foundation of China” has the foremost funding. In further study, it was observed that almost all funding agencies are related to technology.

4.3 Network analysis

Network analysis in bibliometric studies involves examining the connections and relationships between various elements, such as authors, journals, keywords, and institutions within a specific research domain. It helps understand the structure, trends, and collaborations within the scholarly community. VOSviewer provides three categories of Visualization [62]: (i) *Network*, (ii) *Overlay*, and (iii) *Density Visualization*.

Network visualization is one in which items are illustrated with labels and circles by default. An item’s weight regulates the label’s size, and the circle’s size gives an idea of the item’s weight. If the item’s weight is higher, the larger the label, and the larger the circle of the item. The color of the item is identified by which cluster it belongs to. The items are interconnected with each other through links. By default, 1000 links are permissible to be displayed. These 1000 selected links are the strongest among all the links.

Overlay visualization is just like network visualization; the only difference is that the items are colored differently. The score of an item is utilized to set the color in overlay visualization. The colors can be done in two different ways. One is by default, and the other is a user-defined color. If the user has not defined the colors, then by default, colors vary from blue (item having lowest score) to green to yellow (item having highest score).

Density visualization can be of two types: Item Density and Cluster Density. Like Network visualization and Overlay visualization, items are denoted by labels in Item density visualization. The color at every point in item density visualization tells about the density at that particular point. The default selected colors vary from blue to green to yellow. If, at the end, items are very dense, then their color is near yellow, and if items are less dense, then the color is near blue. On the other hand, if items are assigned to clusters, then cluster density visualization is obtained. It is just like density visualization except in the case that the density of each of the clusters is shown separately. For bibliometric analysis, in

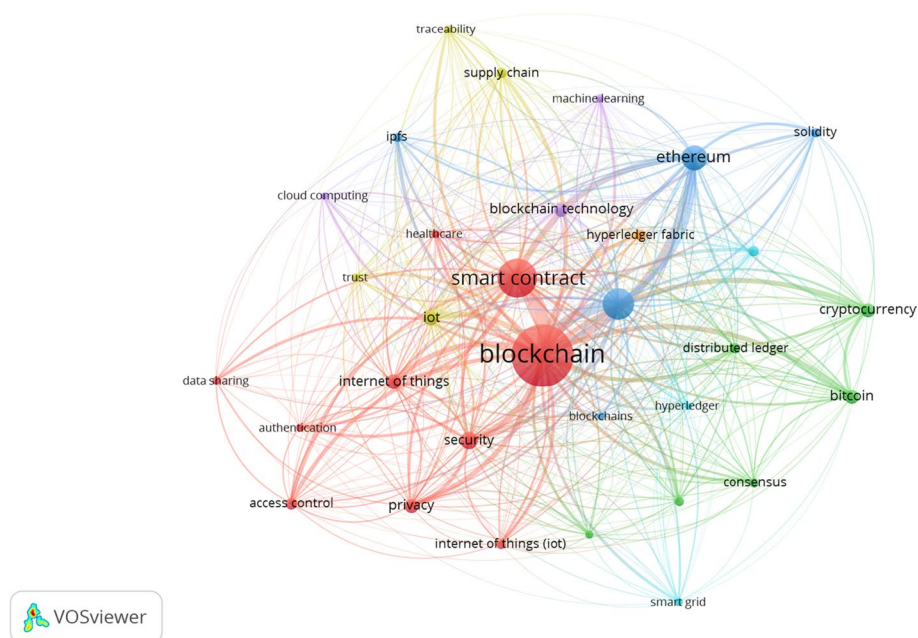


Fig. 5 Co-occurrence analysis of author keywords (Network Visualization created with VOSViewer)

VOSviewer, there are two counting techniques: *complete counting* and *fractional counting* [98].

4.3.1 Co-occurrence analysis of author keywords

In co-occurrence analysis, the intensity of the connection is depicted by how often two unique elements are mentioned together in various publications.

In the co-occurrence analysis of author keywords, the constraint is put on the last occurrence of the keyword. A keyword is included in the study only if it has occurred at least 100 times. By applying this constraint among 13041 keywords, only 32 are included in the analysis. These 32 items are grouped into 7 clusters, as shown in Fig. 5. It can be observed that "blockchain" is the keyword with the largest circle and label size. Blockchain has the highest total link strength of 8444 with 6937 occurrences, followed by "smart contract."

4.3.2 Co-citation analysis of cited references and sources

The link joining two items, such that both are cited in the same document, is a co-citation link. The fractional counting method is utilized to analyze the co-citations. Cited reference analysis of co-citations includes at least 50 cited references. With this threshold, out of 327165 cited references, 27 fulfilled the criteria. For all these 27 cited references, the overall intensity of the co-citation connections with each cited reference will be determined. In total, 26 references

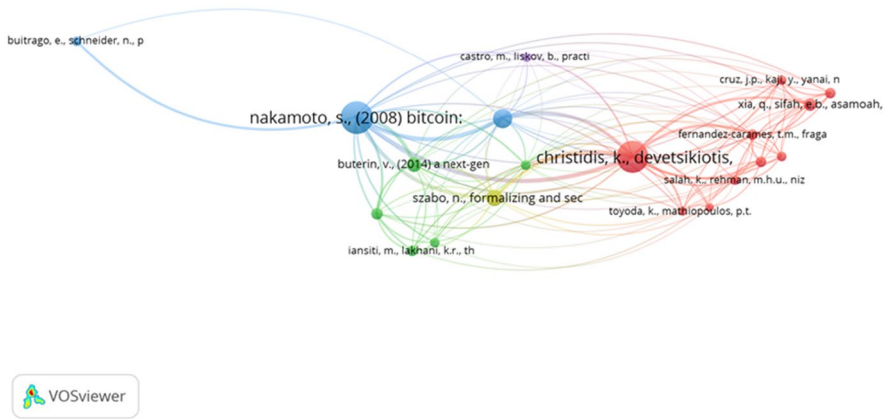


Fig. 6 Co-citation analysis of cited references (Network Visualization created with VOSviewer)

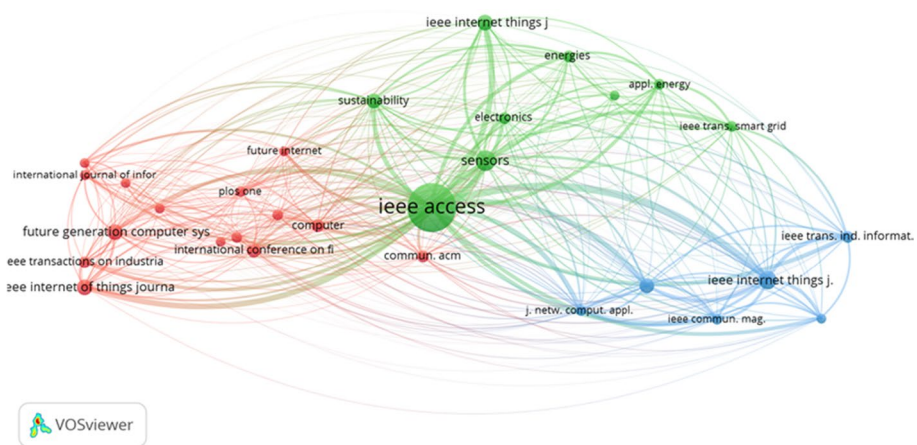


Fig. 7 Co-citation analysis of cited sources (Network Visualization created with VOSviewer)

are connected, as depicted in Fig. 6. Cluster 1 (Red) includes the co-citations of "Blockchains and Smart Contracts for the Internet of Things" [99]. Cluster 2 (Blue) contains the co-citations of Nakamoto white paper, "Bitcoin: a peer-to-peer electronic cash system" [1]. Cluster 3 (Yellow) includes the co-citations of "Formalizing and Securing Relationships on Public Networks" [100]. Cluster 4 (Violet) has co-citations of "Practical Byzantine Fault Tolerance" [101]. The primary reason behind the high co-citation of these articles is that they are the basis of blockchain technology and smart contracts.

Cited source analysis of co-citations excludes the sources having less than 400 citations. With this threshold, out of 125586 sources, 34 fulfilled the criteria. The cumulative co-citation

link strength of these 34 sources with other cited materials will be evaluated. These interactions are illustrated in Fig. 7. "IEEE Access" has the greatest total link strength of 65124 with 9919 citations, followed by "IEEE Internet of Things Journal."

4.4 Research questions

RQ1: What are the hot topics in blockchain-based smart contracts?

The hot topics in Smart Contracts include Ethereum, Stellar, and Hyperledger Fabric, which are prominent for developing smart contracts. Additionally, there is a focus on the theoretical possibilities of artificial intelligence in creating smarter contracts that delegate tasks to machine thinking. Lawyers are exploring whether smart contracts can stand alone as valid contracts under Swiss contract law. Moreover, there is a scientific discussion on enhancing smart contracts' adaptability, responsiveness, and security, particularly in real estate transactions, using blockchain technology and digital records.

RQ2: What is the present state of research for smart contracts development?

The current research on developing Smart Contracts focuses on enhancing performance, scalability, and security. Researchers are exploring new architectures to address existing challenges, such as opacity, tampering, and low efficiency in traditional contracts, by exploiting blockchain's decentralized, tamper-proof, and immutable nature. Additionally, there's an increasing focus on implementing Smart Contracts for various applications, such as the Internet of Things (IoT), healthcare, supply chain management, and business ventures on consortium blockchains. Scholars are also analyzing the security, scalability, and maintainability aspects of Smart Contracts to predict future development prospects and application scenarios. Overall, the research aims to unlock the full potential of Smart Contracts and revolutionize traditional operating procedures across industries.

RQ3: What are the applications of smart contracts?

Smart contracts have diverse applications across various sectors. They enable process automation, eliminate intermediaries, reduce costs, and enhance security. In logistics and supply chain management, blockchain smart contracts ensure product transparency, security, and traceability, enhancing consumer trust and protecting intellectual property rights. Moreover, smart contracts provide an efficient and secure alternative to traditional paper contracts in the insurance industry, improving performance expectancy, technological context, and initial user trust. Overall, blockchain technology's decentralized, peer-to-peer, and tamper-resistant nature, combined with the self-executing capabilities of smart contracts, makes them valuable tools for real-time, cost-effective, and secure task execution across various domains.

RQ4: Which blockchains offer the best smart contract formulations?

Based on the research data, Ethereum stands out for its smart contract development resources, tools, and community support. EOS blockchain excels in execution speed and cost. Stellar is noted for predictability and user-friendly transaction building in smart contract development. On the other hand, the UTXO model, although limiting in some blockchains like Bitcoin, presents a secure alternative by requiring transactions to spend past ones, thus thwarting transaction-ordering attacks. The proposed Theorem-Carrying-Transaction (TCT) technology also aims to enhance smart contract security by focusing on design-level properties and enabling efficient proof checking for safety properties in invoked contracts.

RQ5: What are the challenges and limitations of smart contracts?

Smart contracts face several challenges and limitations. These include language restrictions, infrastructure limitations, a lack of educational resources for developers, security vulnerabilities, immutability issues, and economic implications. Additionally, the proliferation of smart contracts needs to be improved by risks, weaknesses, statutory provisions, security and privacy challenges, growing costs, scalability issues, and lack of adoption. The coding nature of smart contracts is a new dispute requiring innovative dispute resolution mechanisms like Kleros, based on blockchain and game theory. Despite advancements in blockchain technology and smart contracts, these challenges are yet to be solved to enhance efficiency, security, and adoption in decentralized applications.

RQ6: What are the topics for investigation and the potential fields for upcoming research in smart contracts?

Research on smart contracts has highlighted several critical issues and areas for future exploration. Challenges include security threats due to the immutable nature of smart contracts, difficulties in bug detection and vulnerability assessment, and concerns regarding rigidity, uncontrollability, and comprehension by non-experts. Future research directions involve examining practical working models of smart contracts, addressing their limitations, and enhancing their functionality. Additionally, exploring the development trends, application scenarios, technical issues, and potential prospects of smart contract technology is crucial for advancing this field. Overall, the research emphasizes the need for continuous improvement, innovation, and in-depth analysis to harness the full potential of smart contracts in various domains.

5 Practical implications of the study conducted

This study on blockchain for smart contracts has several practical implications for academia and industry:

5.1 Academic implications

1. **Research Trends and Gaps:** The study identified the trends, gaps, and emerging topics within blockchain and smart contracts. It helps researchers focus on unexplored areas and contribute to advancing knowledge.
2. **Citation Patterns:** By understanding the highly cited articles (presented in Table 6), the researchers can identify the most influential works and build upon the existing knowledge.
3. **Collaboration Opportunities:** The co-authorship pattern encourages collaborative research among academicians of different institutes and disciplines.
4. **Methodological Insights:** The paper provides insights into the effective bibliometric research methodology, which helps refine the research practices in this domain.

5.2 Industry implications

1. Innovation and Development: The insights gained from the most cited and influential research guide industry developers in focusing their efforts on promising technologies and approaches.
2. Strategic planning: Organizations can leverage the analysis to understand the competitive landscape, identify the key players, and make informed strategic decisions.

5.3 Cross-sectoral implications

1. Education and Training: Universities and training programs can use bibliometric insights to design curricula that reflect current industry demands and research trends.
2. Investment and Funding: Investors and funding agencies can use bibliometric data to assess research's potential impact and relevance in this domain.

This bibliometric analysis bridges the gap between academia and industry by fostering collaboration, innovation, and informed decision-making across sectors.

6 Limitations of the study

The Bibliometric analysis of “Blockchain-based Smart Contracts” is significantly valuable. However, it also has some limitations that can't be neglected. One major limitation is the time-dependent bias, where metrics like the h-index favor established researchers over younger ones. The other limitation is that the search query to retrieve the publications is limited to 15 January 2023. Over time, the search results may vary due to the continuous influx of studies.

7 Conclusion and future directions

Blockchain Technology is trendy among researchers and industrialists globally. Among its many applications, smart contracts stand out for their utility. These contracts are simple pieces of code that work upon blockchain technology for facilitation, execution, and enforcing the tenure of a contract between untrusted parties. Researchers and developers believe that smart contract development is in its initial stages, and its full potential is yet to be explored. Hence, this research study aims to provide insights to scholars and researchers new to blockchain-based smart contracts. The study uses the bibliometric data extracted from Scopus for the last ten years (2013–2022) under the theme of smart contracts. The publications are screened based on inclusion and exclusion criteria; 10,862 are eligible for the analysis. The focused research questions are” hot topics in smart contracts, the present state of research, use-cases, best smart contract formulations, challenges, limitations, research issues, and future research directions. The study illustrates that researchers are keen to implement smart contracts in domains like the Internet of Things, healthcare, supply chain management, identity management, security, privacy, voting, and higher education. The research on smart contracts is increasing

exponentially. Therefore, as a future work, scholars and researchers can perform bibliometric analysis on individual applications of smart contracts.

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Data availability Data can be provided upon request.

Declarations

Conflict of interest The authors declare no potential conflicts of interest.

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